

Circuit Diagram & Pin Connections

ESP8266 NodeMCU wiring guide for all components

System Block Diagram Description

5V Power Supply	→	ESP8266 NodeMCU	→	MQ-2 Sensor
MQ-2 Sensor (A0)	→	ESP8266 (A0)	→	Threshold Comparison
ESP8266 (D3)	→	Red LED		(Hazard)
ESP8266 (D4)	→	Relay Module	→	Exhaust Fan (230V)
ESP8266 (D5)	→	Buzzer		(Alarm)
ESP8266 (D6)	→	Green LED		(Safe)
230V Power Supply	→	Relay COM/NO	→	Exhaust Fan Motor

Complete Pin Connection Table

Component	Component Pin	ESP8266 Pin	Wire Color (Suggested)	Notes
MQ-2 Sensor	VCC	VU (5V)	Red	Must use 5V not 3.3V
MQ-2 Sensor	GND	GND	Black	Common ground
MQ-2 Sensor	A0 (Analog Out)	A0	Yellow	Analog smoke reading
MQ-2 Sensor	D0 (Digital Out)	Optional D7	White	Optional digital threshold
Relay Module	VCC	VU (5V)	Red	Relay coil power
Relay Module	GND	GND	Black	Common ground
Relay Module	IN	D4	Blue	Active LOW trigger
Relay Module	COM	Fan wire 1	—	Fan circuit common
Relay Module	NO	230V Line	—	Normally Open — fan ON when relay ON
Red LED	Anode (+)	D3 via 220Ω	Red	Hazard indicator
Red LED	Cathode (-)	GND	Black	
Green LED	Anode (+)	D6 via 220Ω	Green	Safe/clean air indicator
Green LED	Cathode (-)	GND	Black	
Orange LED	Anode (+)	D7 via 220Ω	Orange	Warning/threshold indicator
Orange LED	Cathode (-)	GND	Black	

Buzzer	Positive (+)	D5	Purple	Active buzzer — no resistor needed
Buzzer	Negative (–)	GND	Black	

■ ■ Safety Notes

- The exhaust fan (DC Motor) runs on 230V AC mains via the relay — handle with extreme care.
- Always double-check relay wiring before connecting to mains power.
- The ESP8266 operates at 3.3V logic but A0 can read 0–1V (use a voltage divider if MQ-2 output exceeds 1V).
- MQ-2 sensor requires ~20 seconds warm-up time before accurate readings.
- Ensure all GND pins are connected to a common ground.